

Intelligent Manufacturing Engineering, China University of Mining and Technology, Beijing (CUMTB)
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Research Interests

World models and their applications in autonomous driving and embodied intelligence; intelligent manufacturing; machine vision; time-series forecasting; eVTOL and low-altitude autonomous systems.

Education



China University of Mining and Technology, Beijing
B.Eng. in Intelligent Manufacturing Engineering

Sep. 2024 – Jun. 2028 (expected)



Tsinghua University, School of Vehicle and Mobility
Jointly Supervised Undergraduate Researcher, Academician Li Jun's Research Group
Host advisor: Prof. Huang Chaosheng Home advisor: Prof. Zhang Yunjiu

Sep. 2025 – present



Shen Yang No.31 Senior High School
Robotics Club Member

High School

- Built a competition-ready robotics project, sparking an early interest in research; pandemic restrictions prevented national-submission participation.

Research Experience



Time-Series Forecasting and World Models for Autonomous Driving, Academician Li Jun's Research Group, School of Vehicle and Mobility, Tsinghua University Sep. 2025 – present
Role: Jointly Supervised Undergraduate Researcher

- Study time-series forecasting and world-model applications in autonomous driving through trajectory-data processing, literature comparison, experiments, and result analysis.



Project Leader, Undergraduate Research and Innovation Project, Office of Academic Affairs, CUMTB Oct. 2025 – present

“Eye of Demeter – Technology Empowering Rural Revitalization: An Energy-Efficient Chicken Farm Leading a New Chapter in Green Agriculture” (Advisor: Prof. Zhang Yankui)

- Lead proposal design, task allocation, advisory coordination, and milestone management from initiation through execution.

Team Member, Undergraduate Innovation Training Program, CUMTB – May 2026 – Apr. 2027 (expected)
Ivy-Inspired Soft Climbing Robot

Advisor: Prof. Li Yan Role: bio-inspired suction-cup design

- Lead bio-inspired adhesion design and multi-surface testing; support integration of pneumatic, SMA, retraction, and visual-perception modules.



Remote Research Intern, Institute of Automation, Chinese Academy of Sciences May 2025 – Jun. 2025 (Beijing)

- Completed introductory research training through literature review and technical investigation.

Publications and Research Outputs

Published Conference Paper

- Pengyi Xiao, et al.. **Energy Management Method for Diesel Hydrogen Hybrid Vehicles and Evaluation of Renewable Energy Applications**. *Published conference paper*. Co-author.

Under Review

- Pengyi Xiao, et al.. **Title withheld during peer review**. *Renewable and Sustainable Energy Reviews*. Co-author; under review.
- Pengyi Xiao, et al.. **SDRA: A Spatiotemporally Decoupled RL Agent Enhanced by a Vision-Language Model for Heavy-Duty Vehicle Eco-Driving**. *AAAI 2027*. Co-author; under review.

Works in Progress

- One patent application and one software-copyright application are in preparation as outputs of the undergraduate research and innovation project.

Open-Source Contributions

Curator and Contributor, Awesome Autonomous Driving World Model | [GitHub Repository](#)

- Curated papers, codebases, and research resources on autonomous-driving world models for a structured, accessible literature map.

Selected Engineering and Innovation Projects



Intelligent Line-Following Car Competition Project – 1st Overall University-Wide Dec. 2024 – Apr. 2025

Role: Hardware and Debugging Lead

- Led vehicle assembly, module integration, debugging, and fault diagnosis for stable competition performance.

Energy/Water Conservation and Mathematical Modeling Competition Projects – First Prize, Beijing Apr. 2025 – present

Role: Team Leader (Overall Coordination)

- Led topic selection, solution refinement, analysis, presentation, and paper preparation.

Biomimetic Blue Morpho Butterfly – 13th Beijing Undergraduate Mechanical Innovation Design Competition, First Prize (Beijing)

Role: Team Member

2nd CUMTB 3D Printing Creative Design Competition Finals – Third Prize

Dec. 2025

Role: Team Member



Technical Team Member, Gaitly Team – AI Gait Correction and Scenario-Adaptive Modular Baby Walker System, “Challenge Cup” Student Entrepreneurship Competition (UIBE) 2025 – present

- Developed the technical solution; advanced with the team from the UIBE school round.

Awards and Honours

- “Best Crisis Response” Award, Beijing International Model United Nations (BIMUN)
- First Prize, Beijing Energy/Water Conservation and Low-Carbon Emission Reduction Social Practice and Technology Competition
- First Prize (Beijing), National College Students’ Mechanical Design Innovation Competition
- Second Prize (Beijing), Higher Education Cup National Undergraduate Mathematical Contest in Modeling
- Third Prize (Beijing), China International College Students’ Innovation Competition
- Third Prize (Beijing), 16th “CP Cup” National College Students’ Market Survey and Analysis Competition
- Third Prize (Beijing), National College Students’ Financial Challenge Competition
- First Prize, University Line-Following Smart Car Competition (team ranked 1st overall)
- First Prize (College level), Career Planning Competition, School of Mechanical and Electrical Engineering (ranked 1st in college)
- 2025 Advanced Individual in Summer Social Practice; Outstanding Team Leader; First Prize, Outstanding Practice Report
- 2025 University Third-Class Scholarship – Outstanding Student Cadre
- Additional university-level awards (5)

Internship Experience

Research Assistant, China Green Intelligence Urban Research Institute Jul. 2025 – Aug. 2026

Affiliated with the School of Vehicle and Mobility, Tsinghua University

Event Operations Assistant, BMW Brilliance (Huachen BMW) Winter Event Feb. 2025

Part-Time Chinese Language Instructor, Education Institution, Shenyang Nov. 2024 – Jan. 2025

Student Leadership

Vice President, University Employment & Entrepreneurship Association Sep. 2024 – Jun. 2026

President, Model United Nations Association Sep. 2025 – Jun. 2026

Class Monitor, Intelligent Manufacturing Class 24-1 Sep. 2024 – Sep. 2025

Team Leader, Social Practice Team to Beijing, School of Mechanical and Electrical Engineering Jul. 2025 – Aug. 2026 (two consecutive years)

Delegation Head, Beijing International MUN (BIMUN), Northeast China Universities MUN (NIMUN), 2024 – 2025
2025 UIBE MUN

Program Member, 2025 CUMTB (Beijing) New Youth Global Competency Talent Development Program Jun. 2025 – Mar. 2026

Language Proficiency

- CET-4 passed; preparing for IELTS and CET-6. Working proficiency in reading academic literature and written research communication.